2.2 ClusterFLCA Features

2.2.1 Three practical constraints

ClusterFLCA[13] was developed to transform author collaboration networks into interpretable follower–leader clusters by enforcing three practical constraints: each cluster must contain at least three members, each non-top-ranking leader must have at least two followers, and each follower is assigned to only one leader based on the strongest connection[11]. These rules support adequate cluster size, representative leadership, and structurally meaningful collaboration patterns, as shown in Supplemental Appendices 3 and 4.

2.2.2 M**athematical formulation of FLCA**

1. Network definition

Let a weighted undirected network be:

$$G = (V, E, W)$$

- $V = \{v_1, v_2, \dots, v_n\}$: set of nodes
- $E \subseteq V \times V$: set of edges
- $W = \{w_{ij}\}$: edge weights (e.g., WCD)

Each node has an importance score:

$$s_i = \text{value}(v_i)$$

2. Leader–Follower assignment

For each edge (i, j), define direction:

$$(i \to j) = \begin{cases} i \to j, & \text{if } s_i \geq s_j \\ j \to i, & \text{otherwise} \end{cases}$$

Thus:

- Leader = higher score node

- Follower = lower score node

3. Maximum-link selection (core rule)

Each follower j selects **one leader**:

$$L(j) = \arg\max_{i \in \mathcal{N}(j)} \left(w_{ij} + \epsilon s_i\right)$$

where:

- $\mathcal{N}(j)$: neighbors of j
- $\epsilon \ll 1$: small tie-breaking term

This matches the code logic:

slice(1) after arrange(desc(WCD_adj))

4. Cluster assignment (chain merging)

Initialize:

$$c_i^{(0)} = i$$

Iteratively update:

$$c_j^{(t+1)} = \begin{cases} c_{L(j)}^{(t)}, & \text{if } c_j^{(t)} \neq c_{L(j)}^{(t)} \\ c_j^{(t)}, & \text{otherwise} \end{cases}$$

Until convergence:

$$c_j^* = c_{L(j)}^*$$

This is the **chain propagation** seen in the loop:

follower -> leader cluster

5. Leader definition

A node is a leader if:

$$\sum_j \mathbf{1}(L(j) = i) \geq 2$$

Additionally:

$$i^* = \arg\ \max_i\ s_i$$

is always a leader.

6. Minimum cluster constraint

For each cluster C_k:

$$\mid C_k \mid \geq m$$

where:

$$m = 3$$

If violated:

$$C_k \rightarrow C_{parent(k)}$$

Implemented as **bubble-up merging**

7. Objective interpretation (implicit)

FLCA does not optimize a single explicit function but implicitly maximizes:

$$\max \sum_{j \in V} w_{L(j),j}$$

subject to:

- single leader constraint
- hierarchical propagation

- minimum cluster size

8. Silhouette-based validation

Distance:

$$d(i,j) = \begin{cases} \frac{1}{w_{ij}}, & \text{if edge exists} \\ M(1+\delta), & \text{otherwise} \end{cases}$$

Silhouette:

$$s(i) = \frac{b(i) - a(i)}{\max\,(a(i), b(i))}$$

Key insight with 3 constraints:

$$\boxed{\text{FLCA} = \text{Max-link assignment} + \text{Leader hierarchy} + \text{Iterative cluster propagation}}$$

ClusterFLCA enforces three constraints[11]: (1) each cluster must contain at least three members, (2) each cluster leader—excluding the top-ranking leader—must have at least two followers, and (3) each follower is assigned to a single leader based on the strongest connection. These rules ensure that clusters are not only well separated but also structurally interpretable and consistent with hierarchical network organization. Figures 1 to 6 are illustrated from overwhelmed edges(Figure 1) to a clear and concise network(Figures 5 and 6) [7,8,17]via CkysterFKCA[14] on FLCA tab, as shown below

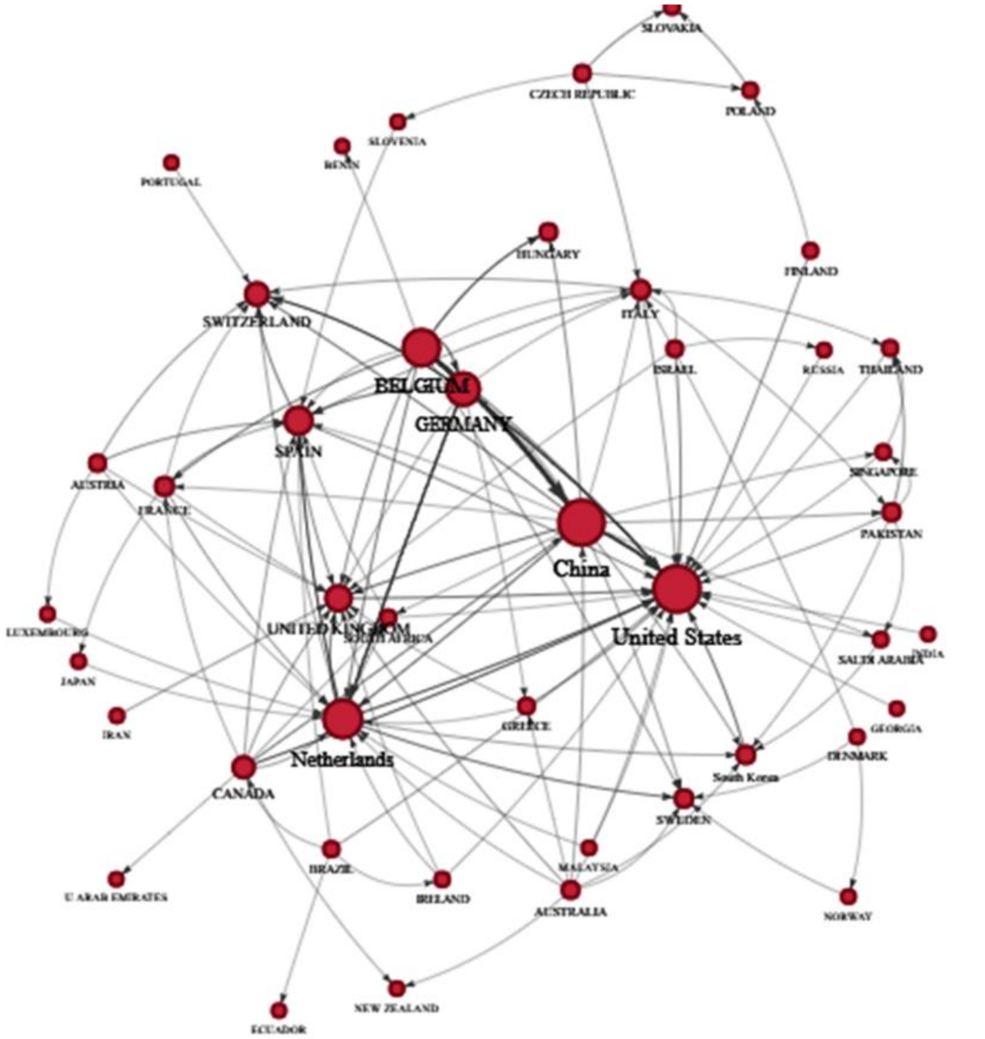

Figure 1 Full metadata edge network before FLCA: Country(too many edges)

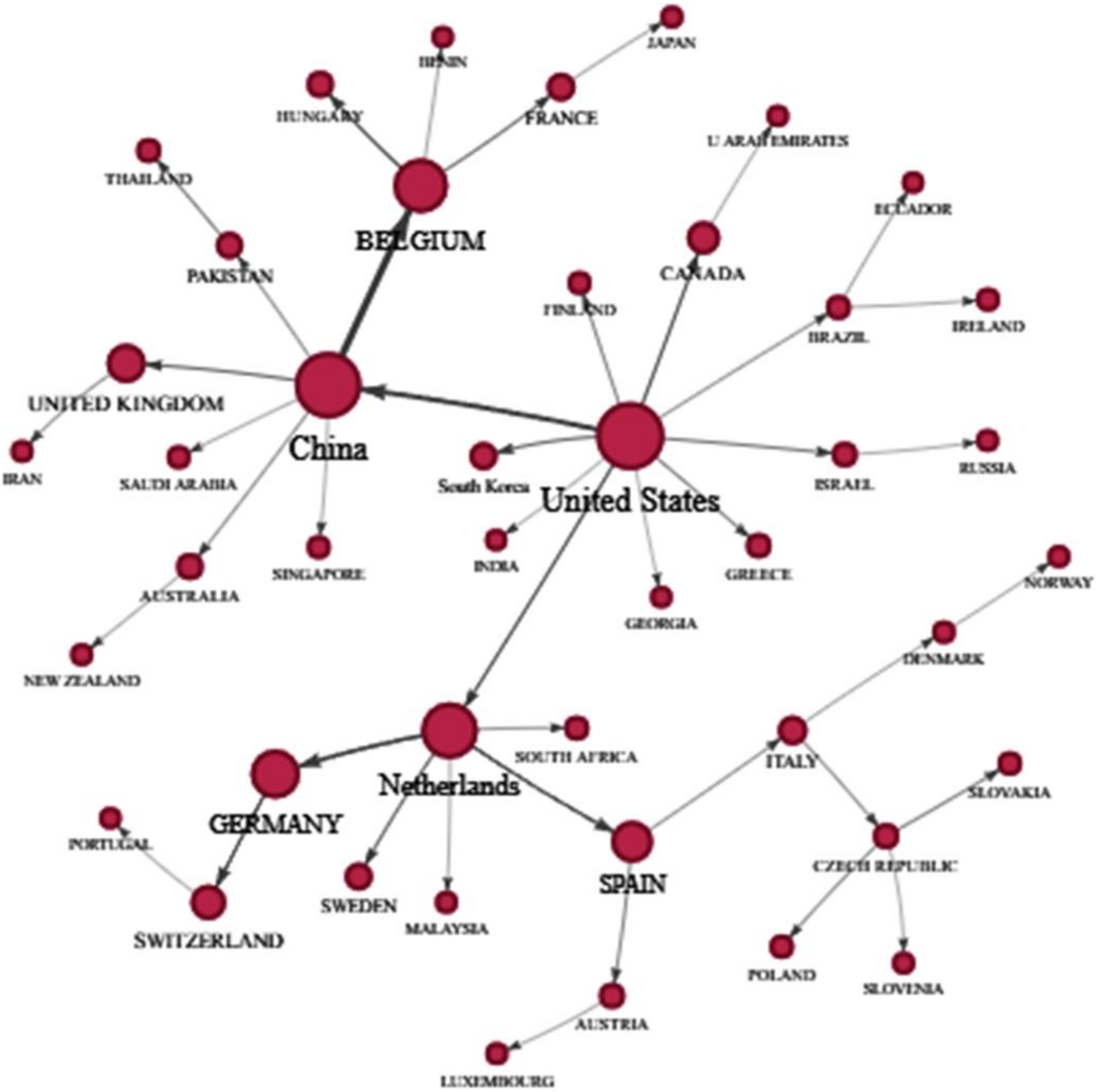

Figure 2 FLCA single-link: one follower to one leader (pre-cluster color with a single edge for followers to the respective leader)

Figure3 FLCA cluster assignment and cluster colors(after FLCA with cluster assignment)

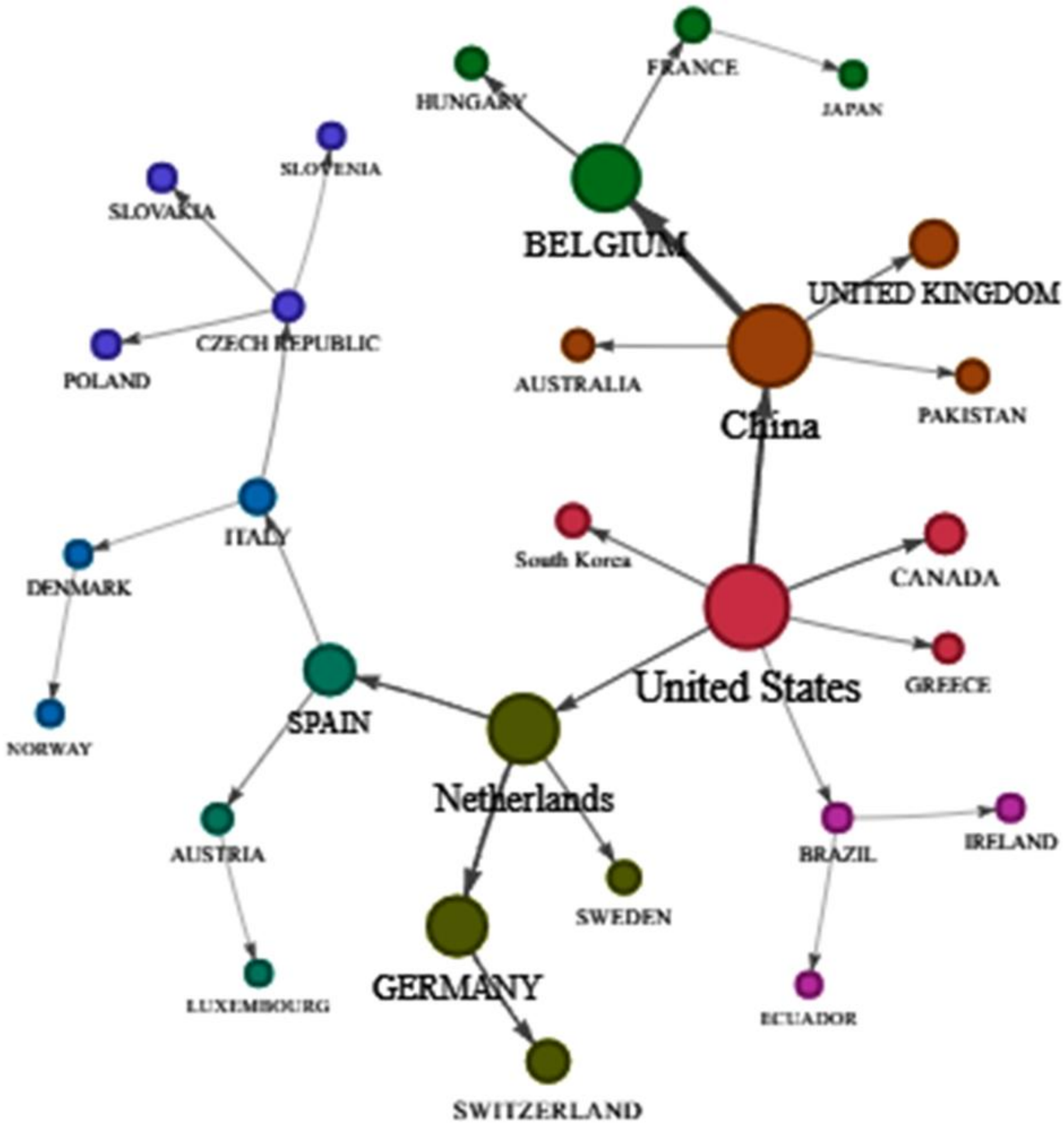

Figure 4 At most 3 nodes retained for each cluster(multi-edges are trimmed beyond 3 conections)

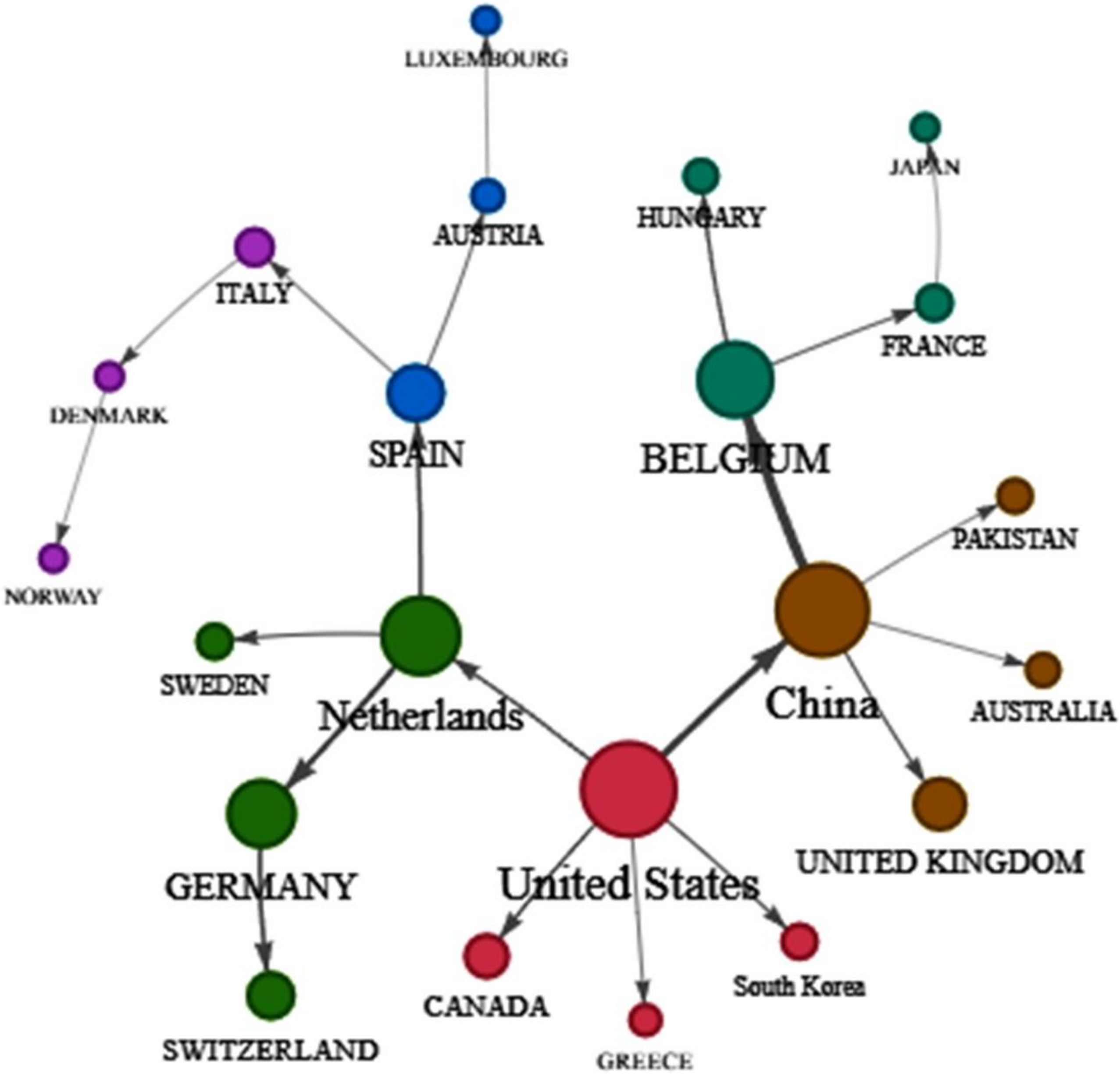

Figure 5 Final Top20+ nodes by major sampling based on larger nodes in clusters[7,8]

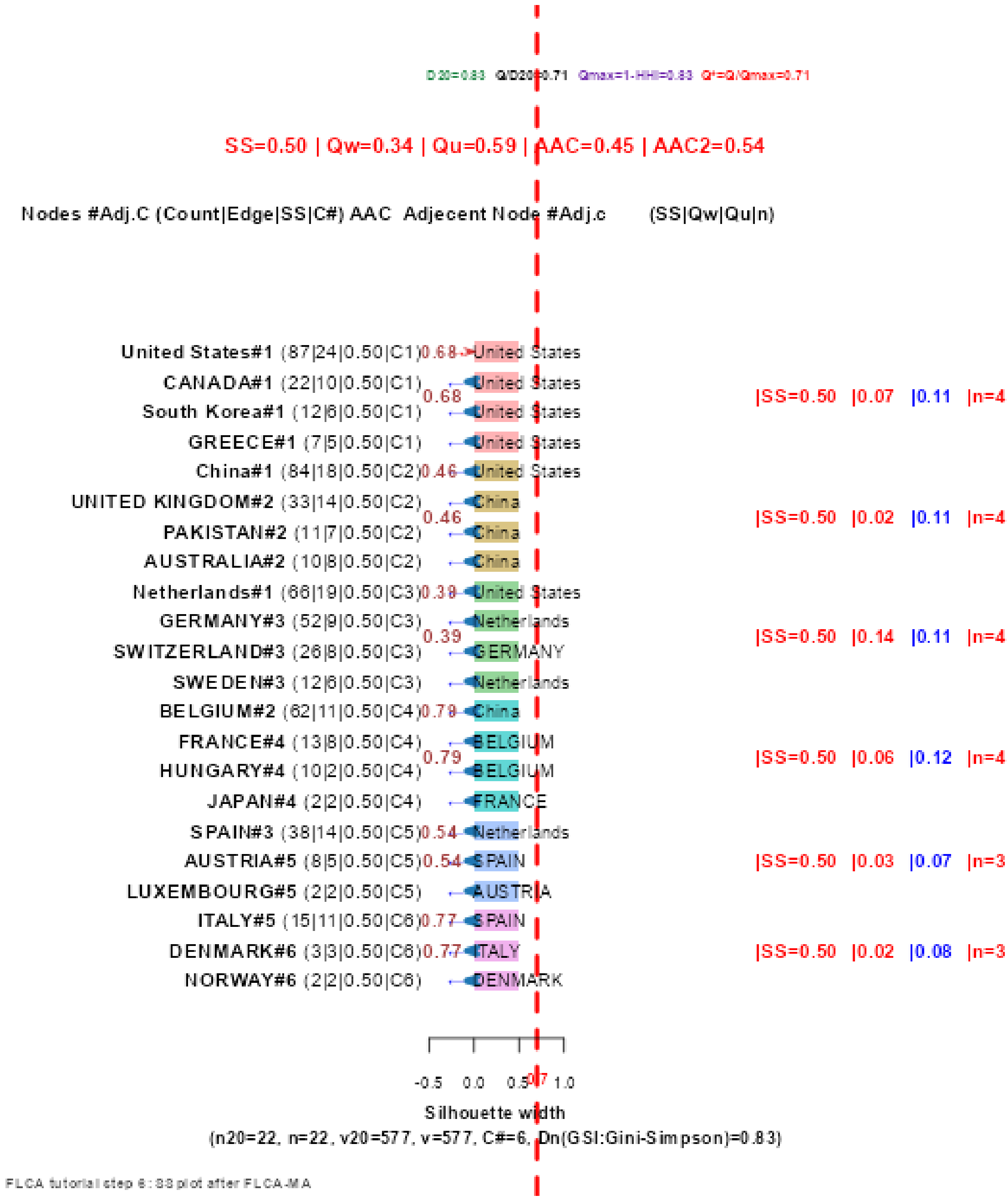

Figure 6 Network with Silhouette score and modularity Q on a single plot[17]

2.3 Two web-based Applications Used in This Study

Two Shiny-based web applications[13,14] were used in this study. The first application, **WoSCited** [13](https://onlinerasch.shinyapps.io/woscited/), supports Web of Science plain-text bibliometric analysis. It allows users to upload or link WoS data, converts records using bibliometrix[15,16], and generates country collaboration networks, keyword co-occurrence networks, cited-reference analyses, thematic maps, and downloadable network data.

The second application, **FLCA Compare[14]** (https://onlinerasch.shinyapps.io/flcacompare/), was developed to compare ClusterFLCA[13] with benchmark clustering algorithms. It accepts nodes and edges data(Supplemental Appendices 1 and 2 which can be downloaded under tab of country collaborations[13]), automatically generates nodes when only edges are provided, applies FLCA-based reduced graph construction, and compares clustering quality using modularity, silhouette, ARI, and NMI. Interactive Network dashboards are used to display full and FLCA-reduced networks, as shown in Graphical Abstract, with adjustable node size, label font, and cluster color.

1. **ARI — Adjusted Rand Index**

   A clustering similarity measure that evaluates how well two partitions agree, **adjusted for chance**. It ranges from **-1 to 1**, where:

   - 1 = perfect agreement
   - 0 = random labeling
   - <0 = worse than random

2. **NMI — Normalized Mutual Information**

   An information-theoretic metric that measures the **shared information between two clusterings**, normalized to account for scale. It ranges from **0 to 1**, where:

   - 1 = perfect match
   - 0 = no mutual information